Final Project: Working with the New York Airbnb Pricing Report in Tableau Public

Estimated time needed: 60 minutes

Required software:

You can complete this project using the free Tableau Public online platform or the Tableau Desktop (Public Edition) app.

About the data set:

Since 2008, guests and hosts have used Airbnb to expand on traveling possibilities and present a more unique, personalized way of experiencing the world. This data set describes the listing activity and metrics in New York city for 2019. This data file includes all the needed information to find out more about hosts, geographical availability, and necessary metrics to make predictions and draw conclusions. This public data set is part of Airbnb.

This public data set is available on the Kaggle website as New York City Airbnb Open Data under the CC0: Public Domain license. The primary data set was retrieved from the website.

Hold the CTRL key while you click to download the following file:

AB_NYC_2019.csv

Attributes of this data set have been explained below:

Column	Type	Description
id	Number	Unique identifier for each Airbnb listing.
host_id	Number	Unique identifier for the host of the listing.
price	Number	Price of the listing per night.
minimum_nights	Number	Minimum number of nights required for a reservation.
number_of_reviews	Number	Total number of views received by the listing.
reviews_per_month	Number	Average number of reviews received by the listing per month.
calculated_host_listing_count	Number	Total number of listings hosted by the same host.
availability_365	Number	Number of days out of the year the listing is available for booking.
latitude	Number	Geographic coordinates of the listing's location.
longitude	Number	Geographic coordinates of the listing's location.
Name	String	Name of the listing.
host_name	String	Name of the host.
neighbourhood_group	String	The broader area surrounding the listing (e.g., Brooklyn, Manhattan).
neighbourhood	String	A specific neighborhood within the city in which the listing is located.
room_type	String	Type of accommodation offered (e.g., Entire Place, Private Room, Shared Room).
last_review	String	Date of the most recent review received by the listing.

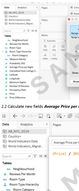
Task descriptions for the project

Task 1. Building hierarchies

1.1 Create a hierarchy named ***Neighbourhood Hierarchy*** using **Neighbourhood** and **Neighbourhood Group**.

1.2 Create a separate hierarchy named ***Room Type Hierarchy*** for different room type entire home/apt, private room and shared room. There is no separate field like *Entire home/apt*, *Private room* or *Shared room*.

First, create a calculated field ***Room Category*** using the **If-then-else** command and then include that in ***Room Type Hierarchy***.



Task 2. Simple data modelling

2.1 Filter the fields **Price**, **Occupancy Rate**, and **Minimum Nights** so that you can work non null values for those fields. You use the **Filter** shelf for filtering in Tableau Desktop (Public Edition).

2.2 Calculate new fields ***Average Price per Night*** by dividing **Price** by **Minimum Nights**.

Task 3: Creating basic visualizations

3.1 Create a Line Chart to visualize the relationship between **Price** and **Reviews Per Month**.

3.2 Create a map using **Neighbourhood Group** and **Neighbourhood**.

3.3 Create a bar chart to compare the previously created field **Average Price per Night** across **Neighbourhood Group**.

Task 4. Creating LOD and complex calculations (by calculated field):

4.1 Create a calculated field named *Price Difference* using LOD calculations to show the difference between a listing's **Price** and the average price in its neighborhood.

4.2 Create a calculated field named ***Total Revenue*** by multiplying **Price** by **Minimum Nights**.

4.3 Create a calculated field named ***Occupancy Rate*** by dividing **Availability 365** by 365.

Task 5. Using tooltips:

5.1 Enhance your map visualizations by adding tooltips that display additional information on hover, such as **Calculated Host Listings Count**, **Price Difference**, **Total Revenue**, and **Occupancy Rate**.

Task 6. Creating a final dashboard:

6.1 Create a dashboard and arrange the created visualizations (map and bar chart) and calculated fields into a visually appealing and informative dashboard.

6.2 Use titles, labels, and annotations to guide viewers through the analysis. Titles, labels and annotations are not mentioned as you can show your creativity. As an example, use the appropriate title in place of **Dashboard 2**.

Task 7. Implementing dashboard actions:

7.1 Create a filter action that allows users to select a specific **Neighbourhood Group** on the bar chart and see the corresponding price distribution and other details in each neighborhood in the map.

7.2 Implement a highlight action that emphasizes listings of a specific **Neighbourhood Group** when selected on the bar chart.

Task 8. Project deliverables:

8.1 Publish your final project workbook and copy its link for submission.

To do this:

1. In the top right corner of the page, click **Publish As** (or **Publish** if you've already published the workbook), then click **Publish As**, name your workbook MyFinalProject(yourname), ensure you check the box **Show sheets as tabs**, and then click **Publish**.
2. In Web Authoring, click **File**, then click **Close**.
3. In the Tableau Public web page for your final project story, click the **Share** icon in the top right toolbar above the final project workbook, and click **Copy Link** in the **Share** dialog box. This will copy the workbook link to the clipboard.

Author(s)

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